

Multiple Choice

1. **Answer:** C

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: The correct answer is true for Statement 1 because it is always possible to define a trivial homomorphism from any group G to another group G' , while Statement 2 is false since a homomorphism can be many-to-one, not necessarily one-to-one.

2. **Answer:** B

Options: A) If A is open, then $A = \mathbb{R}$; B) If A is closed, then $A = \mathbb{R}$; C) If A is uncountable, then $A = \mathbb{R}$; D) If A is uncountable, then A is open.

Note: The correct answer is based on the fact that the set of rational numbers is dense in the real numbers, and if A contains all rational numbers and is closed, it must also contain all irrational numbers, thus making A equal to the entire real line $\mathbb{R}$.

3. **Answer:** C

Options: A) 4; B) 8; C) 12; D) 6

Note: The order of the element $(4, 2)$ in the group $\mathbb{Z}_{12} \times \mathbb{Z}_8$ is determined by the least common multiple of the orders of its components, where the order of 4 in $\mathbb{Z}_{12}$ is 3 and the order of 2 in $\mathbb{Z}_8$ is 4, resulting in a least common multiple of 12.

4. **Answer:** C

Options: A) -11; B) 0; C) 11; D) $33/2$

Note: The correct answer is 11 because the functional equation $f(1+x) = f(x)$ implies that the polynomial f is constant, and since $f(5) = 11$, it follows that $f(x) = 11$ for all x , including at $x = 15/2$.

5. **Answer:** C

Options: A) The ring of complex numbers; B) The ring of integers modulo 11; C) The ring of continuous real-valued functions on $[0, 1]$; D) The ring $\{a + b \cdot \sqrt{2} : a \text{ and } b \text{ are rational numbers}\}$

Note: In the ring of continuous real-valued functions on $[0, 1]$, it is possible for the product of two nonzero functions to be zero at certain points in the interval, demonstrating that the ring contains zero divisors.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The factor group $(\mathbb{Z}_{11} \times \mathbb{Z}_{15}) / \langle (1, 1) \rangle$ is isomorphic to the trivial group because the subgroup $\langle (1, 1) \rangle$ includes all elements of $\mathbb{Z}_{11} \times \mathbb{Z}_{15}$, leading to only the identity element in the quotient.

7. **Answer:** True

Options: A) True; B) False

Note: The element 2 is a generator for the finite field $\mathbb{Z}_{11}$ because its powers modulo 11 produce all non-zero elements of the field, demonstrating that it is a primitive root.

8. **Answer:** True

Options: A) True; B) False

Note: In the ring $\mathbb{Z}_3 \times \mathbb{Z}_3$, every element has an order dividing 3, as the additive identity is reached after adding any element to itself three times, due to the modular arithmetic of $\mathbb{Z}_3$, which confirms that the characteristic is indeed 3.

9. **Answer:** True

Options: A) True; B) False

Note: The dimension of the intersection of two 2-dimensional subspaces of $\mathbb{R}^4$ can be 0, 1, or 2 because the intersection can vary based on the relative positions of the subspaces, with the maximum possible dimension being the minimum of the dimensions of the two subspaces, which is 2, and the minimum dimension being 0 if the subspaces are disjoint.

10. **Answer:** True

Options: A) True; B) False

Note: The statement is true because the minimum value of the expression $x + 4z$, subject to the constraint $x^2 + y^2 + z^2 \leq 2$, can be achieved at the point where $x = -34$ and $z = -34/4$, satisfying both the constraint and yielding the calculated minimum value.

Long Form Response

11. **Answer:** The first ten positive composite integers are 4, 6, 8, 9, 10, 12, 14, 15, 16, 18. The desired quotient is $\frac{4 \cdot 6 \cdot 8 \cdot 9 \cdot 10}{12 \cdot 14 \cdot 15 \cdot 16 \cdot 18}$. After cancellation, we get $\frac{1}{14 \cdot 3} = \frac{1}{42}$.

Note: correct because it accurately identifies the first ten positive composite integers, computes the product of the first five and the next five, and simplifies the resulting fraction through cancellation to arrive at the final answer of $\frac{1}{42}$.

12. **Answer:** If the composition $g \circ f: S \rightarrow U$ is injective, this implies that the function $f: S \rightarrow T$ must also be injective, or one-to-one. To understand why, consider the definition of an injective function: for any two distinct elements s_1 and s_2 in the domain S , if $f(s_1) = f(s_2)$, then it must follow that $g(f(s_1)) = g(f(s_2))$, leading to $g \circ f(s_1) = g \circ f(s_2)$. Since $g \circ f$ is injective, the equality $g(f(s_1)) = g(f(s_2))$ cannot hold unless $s_1 = s_2$. Consequently, if f were not injective, there would exist distinct s_1 and s_2 such that $f(s_1) = f(s_2)$, contradicting the injectivity of $g \circ f$. Thus, for $g \circ f$ to maintain its injective nature, f must indeed be one-to-one.

Note: correct because the injectivity of the composition $g \circ f$ necessitates that distinct inputs in S map to distinct outputs in T via f , ensuring that f must be injective for the overall composition to maintain its injective property.

End of Answer Key